# ESG Rating Dashboard — Project Methodology Report

*Indian Textile & Apparel Manufacturers Peer Set (n = 15) | FY2023-24 / FY2024-25 | Portfolio / Educational Project*

## 1. Project Background

Buyers, investors, and sustainability teams evaluating Indian textile and apparel suppliers face fragmented, inconsistently formatted ESG disclosures (BRSR, GRI, TCFD, sustainability reports, integrated annual reports), making like-for-like comparison difficult. Commercial third-party ESG ratings are also frequently opaque, making it hard to explain why one supplier scores higher than another. This project was built to demonstrate — end to end — how an analyst can construct a transparent, defensible, peer-relative ESG scoring model directly from company-disclosed data, where every score traces back to one visible formula and one disclosed data point.

## 2. Dataset

- **Source:** Company-disclosed data only — BRSR filings, sustainability reports, and integrated annual reports for each company's most recent reporting year (FY2023-24 or FY2024-25). No third-party verification beyond each company's own disclosed CDP/assurance status.
- **Number of companies:** 15 Indian textile & apparel manufacturers (e.g., Shahi Exports, Vardhman Textiles, Arvind Limited, Trident Limited, K.P.R. Mill, Page Industries, Indo Count, Pearl Global, Welspun Living, Siyaram's Silk Mills, Jindal Worldwide, Grasim Industries, RSWM Limited, Gokaldas Exports, Vamani Overseas).
- **Main ESG metrics:** Scope 1 & 2 emissions, renewable energy %, total energy, solar capacity, total water, waste (total/hazardous/plastic), sustainable materials %, gender diversity, SBTi status, CDP score, ISO 14001/GOTS/RE100 certification, ownership type, reporting framework(s) used.
- **Data fields:** 48 raw and derived columns per company spanning company profile, workforce, emissions, energy, water, waste, certifications, climate targets (free text), and supply-chain context (major buyers, raw materials, sourcing geography).

## 3. Data Preparation

*No Power Query (M) steps exist in this workbook — all preparation is done with live Excel formulas so every transformation stays visible.* Key transformations detected:

- Male %/Female % computed on the fly from headcounts (`=ROUND((Count/Total)*100,2)`), not hardcoded.
- Employee & Energy Carbon Intensity derived (`Scope1+2 ÷ Employees`; `Scope1+2 ÷ Total Energy`) with `IFERROR` guards against missing denominators.
- Free-text 'Reporting Standard' field parsed into a structured count via `SUMPRODUCT(--ISNUMBER(SEARCH({"GRI","SASB","TCFD","BRSR","UNGC",...}, text)))`.
- Every qualitative field (SBTi status, RE100, CDP score, ownership) is explicitly branched for 'Not Disclosed'/'Not found' so missing data scores 0 or is excluded from the coverage count — never silently dropped or blank-averaged.
- Peer-relative min-max normalization applied to the three lower-is-better intensity metrics, scaled against this 15-company set specifically.

## 4. Data Modeling

The workbook uses a single-subject, denormalized worksheet design rather than a relational Power Pivot/star-schema model: **Framework_Overview** (methodology documentation) feeds context for **LS_Main_Dataset** (15 rows × 48 columns of raw/lightly-derived data), which is referenced row-by-row by the hidden **Calculations** sheet (metric scores → dimension scores → Overall ESG Score → Rating → Confidence), which in turn populates the presentation-ready **ESG_Scorecard** sheet. Sheets are linked by direct cell references aligned 1:1 by company row — there are no defined table relationships, no fact/dimension split, and no many-to-many joins. This is an appropriate, simple design at n = 15 rows, but is a deliberate simplification relative to a true BI relational model (see Limitations).

## 5. Calculation Methodology (Native Excel Formulas — Not DAX)

This workbook does not use DAX or Power Pivot measures. All scoring logic is native Excel formulas in the **Calculations** sheet, explained here in business terms:

- **Metric score:** each disclosed value is converted to 0–100. 'Higher-is-better' metrics (renewable energy %, sustainable materials %, gender diversity) use the disclosed value directly. 'Lower-is-better' intensity metrics (emissions, water, waste per employee) use peer min-max normalization: `(MAX(peers)-value)/(MAX(peers)-MIN(peers))×100`.
- **Qualitative tier lookups:** nested IF formulas convert categorical disclosures to scores — e.g., SBTi Committed=100, Not Committed=40, Not a member=15, Not found=10, Not Disclosed=0; Ownership Public Listed=100, Private=55; CDP grade A=100 down to D=25.
- **Dimension score** = simple `AVERAGE()` of that dimension's metric scores (no sub-weighting).
- **Overall ESG Score** = `Environmental×0.40 + Social×0.25 + Governance×0.20 + Transparency×0.15`.
- **Rating band**: nested IF mapping the Overall ESG Score to AAA (≥90) through CCC (<40).
- **Data Coverage % / Confidence:** `COUNT()` of usable values across the 12 scored metrics ÷ 12, banded High (≥80%)/Medium (60–79.9%)/Low (<60%) — reported alongside, not blended into, the score.

## 6. Dashboard Design

- **KPI cards:** Overall ESG Score/Rating badge, Environmental/Social/Governance scores, and operational tiles (Solar Capacity, Total Waste, Total Energy, Energy Carbon Intensity, Total Water) give an at-a-glance read before any chart is examined.
- **Charts:** stacked bar for Scope 1 vs. Scope 2 emissions split; donut charts for Renewable Energy % and Sustainable Materials %; horizontal color-graded bar chart ranking all 15 companies; donut chart of ESG rating distribution; bar chart of certification adoption rates across the peer set.
- **Slicers/selection:** a company selector drives the entire Company Profile page from one input, avoiding duplicated layouts per company.
- **Navigation:** two purpose-built views — a single-company profile page (account-manager / supplier-review use case) and a portfolio/peer-ranking page (portfolio-manager / sourcing-lead use case).
- **Visual choices & UX:** green/red conditional coloring on certification badges for instant scan-ability; consistent color grading (red→green) on the rating scale and leaderboard bar chart; a persistent data-source/reporting-year footer for credibility and traceability.

## 7. ESG Scoring Methodology (As Implemented)

Environmental (40% weight, 7 metrics), Social (25%, 2 metrics), Governance (20%, 1 metric), Transparency (15%, 2 metrics) — each dimension score is a plain, unweighted average of its metric scores; there is no metric-level or criterion-level weighting beyond the four dimension weights above. The Overall ESG Score is a straight weighted sum of the four dimension scores, mapped to a 7-band letter rating (AAA down to CCC). A separate Data Coverage %/Confidence flag (High/Medium/Low) is calculated but never blended into the score itself — this keeps 'how good is the company' and 'how much evidence do we have' as two distinct, independently interpretable numbers, which is a deliberate and defensible design choice for a disclosure-based rating.

## 8. Business Insights

- Portfolio average Overall ESG Score is ~57–58/100 (BB/BBB band) — solid disclosure, meaningful room for improvement.
- Page Industries and Welspun Living lead the peer set (high-70s scores, 'A' rating), driven by strong Social/Transparency performance.
- Grasim Industries scores lowest (CCC, high-30s), driven primarily by weak Environmental performance at scale.
- Governance scores cluster near the ceiling (100 for Public Listed, 55 for Private) — Governance barely differentiates companies here, a known limitation of a single-metric proxy.
- Certification adoption is uneven: ISO 14001 (13/15) and GOTS (12/15) are near-universal; SBTi commitment (4/15) and CDP disclosure quality remain real differentiators; zero companies are RE100 members in this peer set.

## 9. Limitations

- Peer-relative scoring only — not comparable to companies outside this 15-company set.
- Governance scored on a single proxy (ownership type) due to no other governance fields being disclosed in source data.
- Farmer/Smallholder Engagement disclosed by only 3 of 15 companies; most score 0 due to genuine non-disclosure, not estimation.
- Single-year, cross-sectional snapshot — no trend/time-series analysis yet.
- No true relational data model (Power Pivot/DAX) — a linear worksheet-formula chain instead, sufficient at n = 15 but not built for scale.
- Educational/portfolio project — not an investment research product; not suitable for investment or procurement decisions.

## 10. Future Scope

- Migrate the calculation engine to Power BI (Power Query for ingestion + DAX measures) for true calculated measures and a relational model.
- Live database or API connection with automated, scheduled refresh instead of manual disclosure updates.
- Deploy to Power BI Service (or an equivalent hosted BI tool) for shareable, no-download access and row-level security by user role.
- SQL-backed staging layer for the raw disclosure data ahead of the scoring engine.
- External benchmarking against a global or sector ESG index, not only peer-relative normalization.
- Predictive/scenario layer — e.g., estimating the score impact of a renewable-energy or waste-reduction target.

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Report generated as part of a Data Analytics portfolio project. All figures reflect the ESG_Dashboard.xlsx workbook (Framework_Overview, LS_Main_Dataset, Calculations, ESG_Scorecard sheets).